Problem Set 4 — Motion & Graphs

Physics 2026–27 · due at the start of class, Tuesday, September 22

NAME ____________________ DATE __________

How to work this set. Every quantitative problem runs the protocol — labeled **K / D / E / A**, units traveling with the math. Signs are physics now, not decoration: a negative Δx or v is a direction, and full credit requires saying which direction. Definitions this week: $\Delta x = x_f - x_i$, $v = \Delta x/\Delta t$, $a = \Delta v/\Delta t$.

1 Displacement vs. distance

Δx

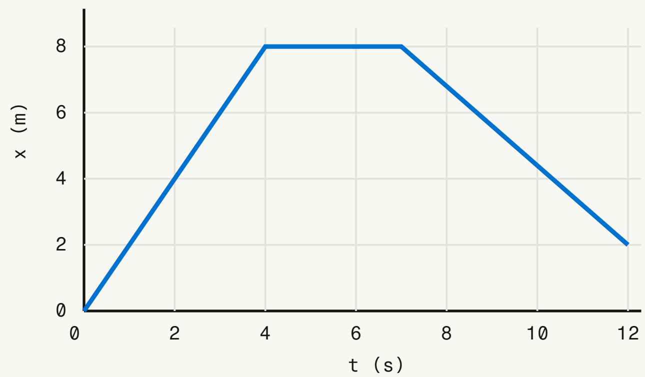
a. A cyclist rides from $x_i = -4$ m to $x_f = +10$ m. Find Δx, and state which direction she moved.

b. A ball rolls from $x_i = 6$ m to $x_f = -2$ m. Find Δx. Did it cross the origin?

c. You walk 8 m to the right, then 3 m back to the left. What distance did you travel, and what is your displacement? One sentence: why are they different?

2 Read the x–t graph

GRAPHS



The graph shows a student walking in front of a motion sensor. Segment I runs 0–4 s, segment II runs 4–7 s, segment III runs 7–12 s.

a. Find the velocity in each of the three segments (KDEA on at least one; show the slope arithmetic on all three).

b. What is the total displacement for the whole 12 s? The total distance traveled?

c. In which segment is the student moving fastest? How do you know from the *shape* alone?

WORK

3 Read the v–t story

GRAPHS

A car's v–t graph shows: it accelerates uniformly from rest to `20 m/s` in 5.0 s (segment I), holds `20 m/s` for 8.0 s (segment II), then brakes uniformly to a stop in 4.0 s (segment III).

a. Sketch the v–t graph, labeling the three segments.

b. Find the acceleration in each segment.

c. During segment II the line is flat. Is the car stopped? Explain what a flat line means on a v–t graph vs. an x–t graph.

SKETCH + WORK

4 Velocity from data

KDEA

A student walks away from a motion sensor in a straight line, from $x_i = 0.8$ m at $t_i = 0$ s to $x_f = 3.2$ m at $t_f = 3.0$ s. Run KDEA to find her velocity.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	DEFINITION
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Acceleration from data

KDEA

A car merges onto I-95, speeding up from 12 m/s to 30 m/s in 6.0 s. Run KDEA to find its acceleration — and check that your final unit is the one acceleration demands.

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6 Negative ≠ slowing down

SIGNS

A delivery truck is backing up (moving in the negative direction) at −3 m/s, and one second later it is moving at −5 m/s.

a. What is the truck's acceleration, sign included?

b. Is the truck speeding up or slowing down? Explain using the signs of v and a — not the word "negative."

7 Draw the story

SKETCHING

Sketch a single x–t graph for this trip, labeling each piece: you leave home walking forward at a steady, brisk pace for 5 s · you stop at the corner for 4 s · you walk back toward home at a steady, *slower* pace for 8 s (you don't quite make it back). Mark where velocity is positive, zero, and negative — and make the return slope visibly gentler than the outbound slope.

SKETCH

Graded on the homework 4–3–2–1 scale. Signs are part of every answer this unit — a velocity without a sign is a speed, and that's only half the story.